Chapter - 0

<- What is Programming ->

What is Programming?

Just like we use Hindi or English to communicate with each other, we use a programming language like python to communicate with the computer.

Programming is a way to instruct the computer to perform various tasks. These instructions, are known as code, are written in programming languages that computer understand and executes to carry out various operations.

What is Python?

Python is a high-level programming language kown for its simplicity and readability.

Note: Python was created by Guido von Rossum in 1991.

Feature of Python

Easy to understand = Less development time
Free and open source
High level language
Portable -> works on Linux / windows / Mac

Installation

Python can be easily installed from Python.org
When you click on the dowload button, Python can be installed right after you complete the setup by excuting the file for your platform.

Chapter - 1

← Modules, Comments & pip →

Lets write our very first python program.
Create a file called hello.py and paste the below code in it

```
Print("hello World")
```
→ Print is a function

Execute this file (.py) by typing python hello.py and you will see Hello wolld printed on the Screen

Modules

A module is a file containing code written by somebody else (usually) which can be imported and used in our programs.

Type of modules

There are two types of module in Python

1- Built in modules → Pre installed in Python
2- External modules → Need to install using pip.

Using Python as a Calculator

We can use python as a calculator by typing "python" + ↵ on the terminal. → This opens REPL or read Evaluate Print loop

Comments

Comments are used to write something which the Programmer does not want to execute.
↳ can be used to mark author name, data etc.

Types of Comments

There are two types of comment in Python

1- Single line comments → writting using #
2- Multiline comments → writting using '''comment'''

Chapter -1 Practic Set

1. Write a Program to print Twinkle Twinkle little star poem in python.

2. Use REPL and Print the table of 5 using it.

3. Install an externall module and use it to perform an operation of your interest.

4. Write a python program to print the contents of a directory using OS module. Search online for the function which does that

5. Label the program written in Problem 4 with comments.

File Edit Selection View Go Run ... Chapter-1

EXPLORER

CHAPTER-1
- .ipynb_checkpoints
- .venv
- first.py
- module.py
- practiceset1.ipynb
- screen.jpg

OUTLINE
TIMELINE

module.py > ...

```python
import pyjokes
#print("printig jokes") this is single line commen
joke=pyjokes.get_joke()
print(joke)
"""
this is multi
line comment
"""
```

first.py | practiceset1.ipynb

practiceset1.ipynb > # Q-1 write a program to print twinkle twinkle

Generate + Code + Markdown | Run All ... Select Kernel

```python
# Q-3 install an external module and
# use it to perform an operation of your interest
import pyttsx3
engine = pyttsx3.init()
engine.say("hello i am learning python ")
engine.runAndWait()
```

[4] Python

```python
# Q-4 write a python program to print the content
# directory using OS module search online for the
# function which does that
import os
directory_path='/'
contents=os.listdir(directory_path)
for item in contents:
    print(item)
```

[6] Python

```
sbin.usr-is-merged
var
usr
lib
home
boot
proc
mnt
lost+found
```

0 files and 30 cells to analyze | Spaces: 4 | Cell 1 of 5

DAT Linux 15:48